Posibilidades Code Part 3

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Matching districts to VTDs

This code document matches Texas legislative districts from the 2011 redistricting cycle, 2021 redistristing cycle, and 2025 u.s. congressional districts to Texas VTD's.

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.0     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
library(geomander)

`%nin%` <- negate(`%in%`)
```

```r
sf24vtd <- read_sf("vtd24sf\\VTDs_24PG.shp")


sf24vtd <- st_transform(sf24vtd, crs = st_crs("EPSG:4326"))


sf24vtd <- st_make_valid(sf24vtd)
```

```r
c_districts_2025 <- read_sf("district_maps/PLANC2333/PLANC2333.shp")

c_districts_2025_matches <- geo_match(
  sf24vtd,
  c_districts_2025,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk <- data.frame(c_districts_2025_matches)

rm(c_districts_2025, c_districts_2025_matches)

gc()
```

```
           used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells  1638986 87.6    3165428 169.1  3165428 169.1
Vcells 11924940 91.0   42089746 321.2 52612136 401.4
```

```r
h_districts_2021 <- read_sf("district_maps/PLANH2316/PLANH2316.shp")

h_districts_2021_matches <- geo_match(
  sf24vtd,
  h_districts_2021,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk$h_districts_2021_matches <- h_districts_2021_matches


rm(h_districts_2021, h_districts_2021_matches)

gc()
```

```
           used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells  1639330 87.6    3165428 169.1  3165428 169.1
```

```
Vcells 11930853 91.1   42089746 321.2 52612136 401.4
```

```
s_districts_2021 <- read_sf("district_maps/PLANS2168/PLANS2168.shp")

s_districts_2021_matches <- geo_match(
  sf24vtd,
  s_districts_2021,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk$s_districts_2021_matches <- s_districts_2021_matches


rm(s_districts_2021, s_districts_2021_matches)

gc()
```

```
            used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells   1638995 87.6    3165428 169.1  3165428 169.1
Vcells  11935123 91.1   42089746 321.2 52612136 401.4
```

```
c_districts_2021 <- read_sf("district_maps/PLANC2193/PLANC2193.shp")

c_districts_2021_matches <- geo_match(
  sf24vtd,
  c_districts_2021,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk$c_districts_2021_matches <- c_districts_2021_matches


rm(c_districts_2021, c_districts_2021_matches)

gc()
```

```
            used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells   1639352 87.6    3165428 169.1  3165428 169.1
Vcells  11940540 91.1   42089746 321.2 52612136 401.4
```

```
h_districts_2011 <- read_sf("district_maps/planh358_shapefile/PLANH358/PLANH358.shp")
```

```r
h_districts_2011_matches <- geo_match(
  sf24vtd,
  h_districts_2011,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk$h_districts_2011_matches <- h_districts_2011_matches


rm(h_districts_2011, h_districts_2011_matches)

gc()
```

```
           used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells  1639233 87.6    3165428 169.1  3165428 169.1
Vcells 11945469 91.2   42089746 321.2 52612136 401.4
```

```r
s_districts_2011 <- read_sf("district_maps/plans172_shapefile/PLANS172/PLANS172.shp")

s_districts_2011_matches <- geo_match(
  sf24vtd,
  s_districts_2011,
  method = "area",
  tiebreaker = T
)

vtd_district_crosswalk$s_districts_2011_matches <- s_districts_2011_matches


rm(s_districts_2011, s_districts_2011_matches)

gc()
```

```
           used (Mb) gc trigger  (Mb) max used  (Mb)
Ncells  1639252 87.6    3165428 169.1  3165428 169.1
Vcells 11950327 91.2   42089746 321.2 52612136 401.4
```

```r
c_districts_2011 <- read_sf("district_maps/planc235_shapefile/PLANC235/PLANC235.shp")

c_districts_2011_matches <- geo_match(
  sf24vtd,
  c_districts_2011,
  method = "area",
```

```r
  tiebreaker = T
)

vtd_district_crosswalk$c_districts_2011_matches <- c_districts_2011_matches


rm(c_districts_2011, c_districts_2011_matches)

gc()
```

```
           used  (Mb) gc trigger  (Mb) max used  (Mb)
Ncells  1639432  87.6    3165428 169.1  3165428 169.1
Vcells 11955449  91.3   42089746 321.2 52612136 401.4
```

```r
write.csv(
  vtd_district_crosswalk,
  "vtd_district_crosswalk.csv",
  row.names = FALSE
)
```